# PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY-DINDIGUL

DEPARTMENT OF INFORMATION TECHNOLOGY

# E-Fir hub: AI-Driven Solutions for Flawless FIR Writing

## IT2891-PROJECT WORK (SECOND REVIEW)

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# Abstract


• E-FIR Hub is an AI-driven system designed to modernize the traditional FIR registration process, which is often time-consuming, error-prone, and challenging for citizens with limited legal knowledge.

• The system allows users to submit complaints through multiple formats such as text, voice, or uploaded documents. Speech-to-Text and OCR technologies convert these inputs into a unified and processable text format.

• Using Natural Language Processing (NLP), the system analyzes complaint data, extracts key information, classifies the type of crime, and intelligently maps it to relevant legal sections for accurate FIR preparation.

• The platform automatically generates structured FIR drafts and supports a verification process by officials, reducing manual effort, minimizing documentation errors, and improving efficiency, transparency, and accessibility in FIR registration.

# Project Objective

**Title:**

E-FIR Hub: AI-Driven FIR Writing System

**Objective:**

To simplify and automate the FIR filing process using AI technologies

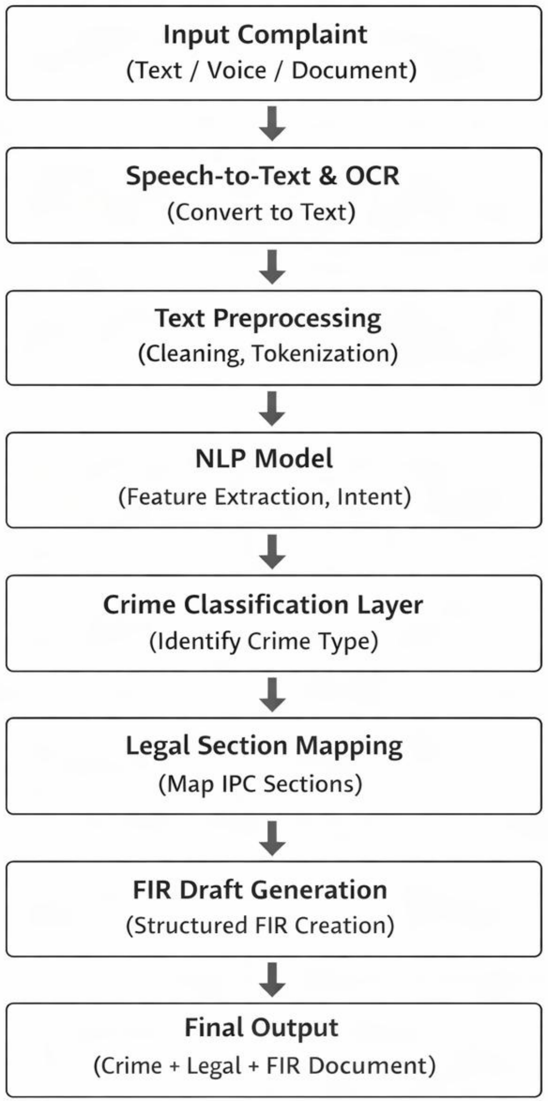
**Overview:**

- Allows complaint submission through text, voice, and documents
- Converts audio and images into text using Speech-to-Text and OCR
- Uses NLP to analyze complaints and classify crime types
- Automatically generates structured FIR drafts
- Suggests relevant legal sections for accuracy

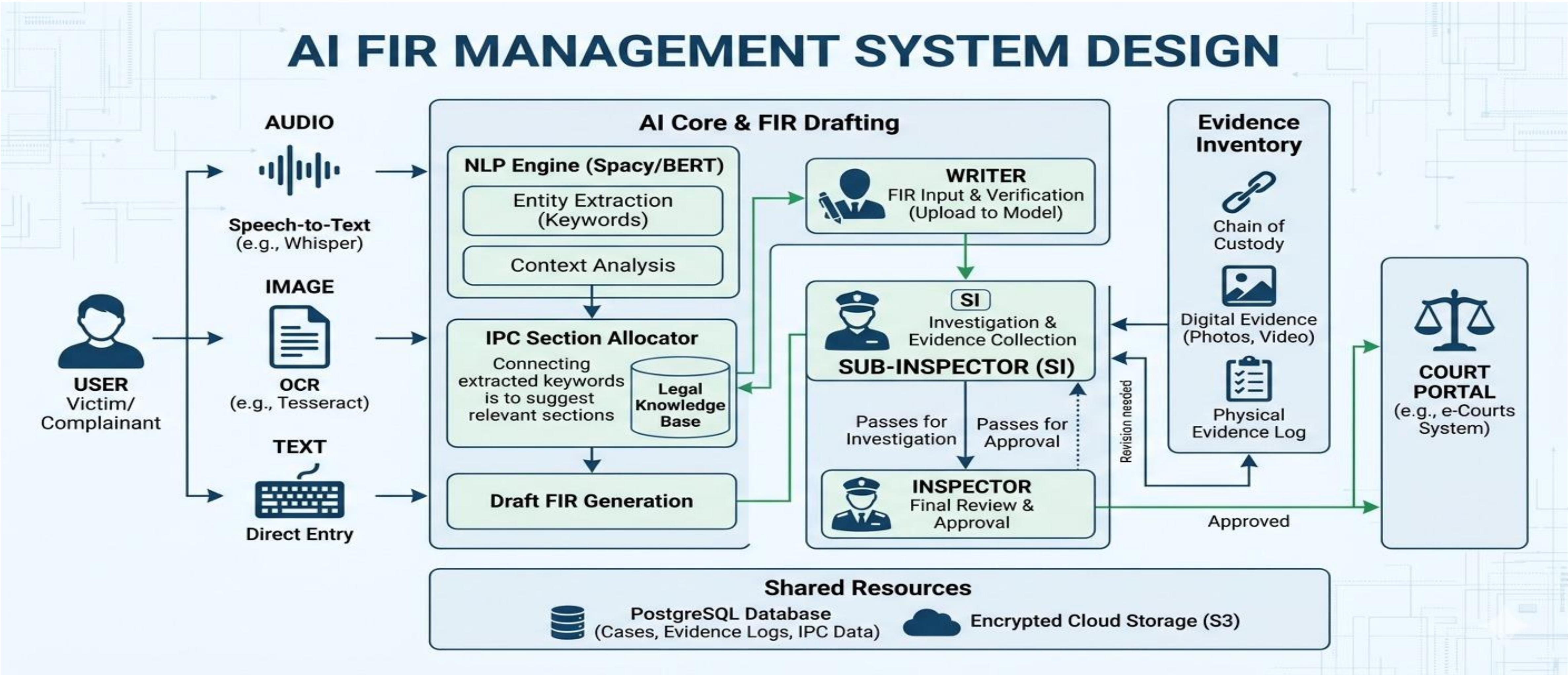
**Outcome:**

- Faster and efficient FIR registration
- Reduced manual errors and paperwork
- Improved accessibility for citizens
- Supports police with quick and accurate documentation

# System Architecture



# System Design



# Module 1 – Data Loader(E-FIR HUB)

**Purpose:**

Preprocess complaint data (text, audio, document)

**Functions:**

- Text input processing
- Speech-to-Text conversion
- OCR for documents

**Processing:**

- Cleaning & normalization
- Tokenization

**Tools:**

NLP, Speech-to-Text, OCR, Python

# Module 2 - Model Builder

**Purpose:**

Build AI model for crime classification and FIR generation

**Functions:**

- NLP-based complaint analysis
- Crime classification
- Legal section mapping

**Process:**

- Feature extraction
- Model training & validation

**Tools:**

Python, NLP Models, Machine Learning

# Module 3 - Training

**Purpose:**

Train AI model for crime classification and FIR generation

**Training Setup:**

- Dataset: Complaint records (text/audio/document)
- Input: Processed complaint text
- Output: Crime category + legal sections

**Process:**

- Data preprocessing
- NLP model training
- Model evaluation (Accuracy, Precision, Recall)

**Workflow:**

Data → NLP Model → Classification → Evaluation → Save Model

# Module 4 - Prediction

**Purpose:**

Predict crime type and generate FIR details from complaint

**Functions:**

- Analyze input complaint (text/audio/document)
- Predict crime category using trained NLP model
- Suggest relevant legal sections

**Process:**

- Input → Preprocessing → Model Prediction
- Map prediction to legal sections

**Output:**

- Crime type + IPC sections
- Generated FIR draft

# Algorithms

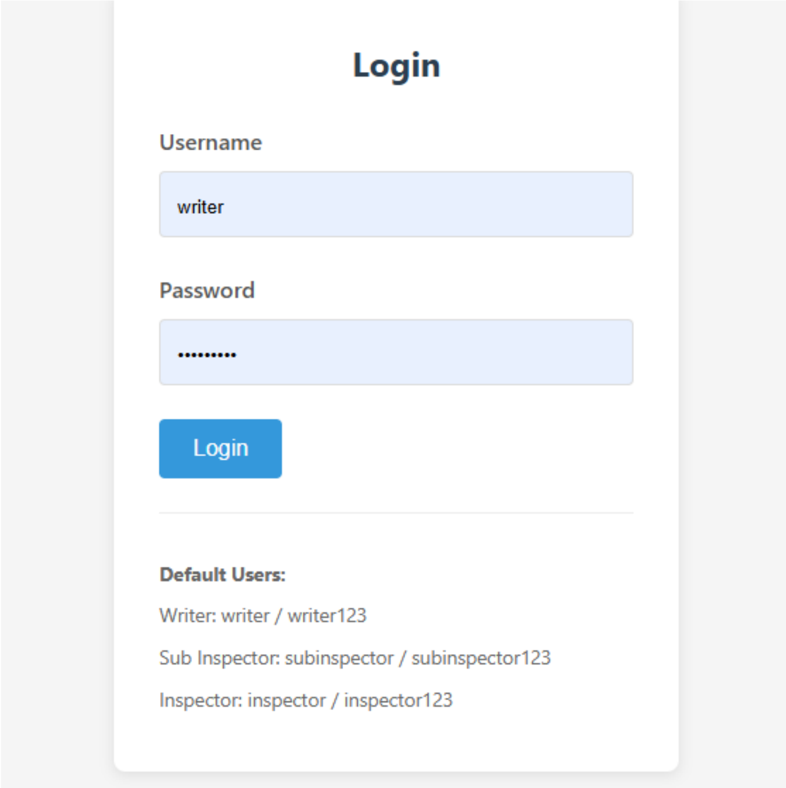
**1. NLP-Based Crime Classification**
- Analyzes complaint text
- Extracts keywords and intent
- Classifies crime type

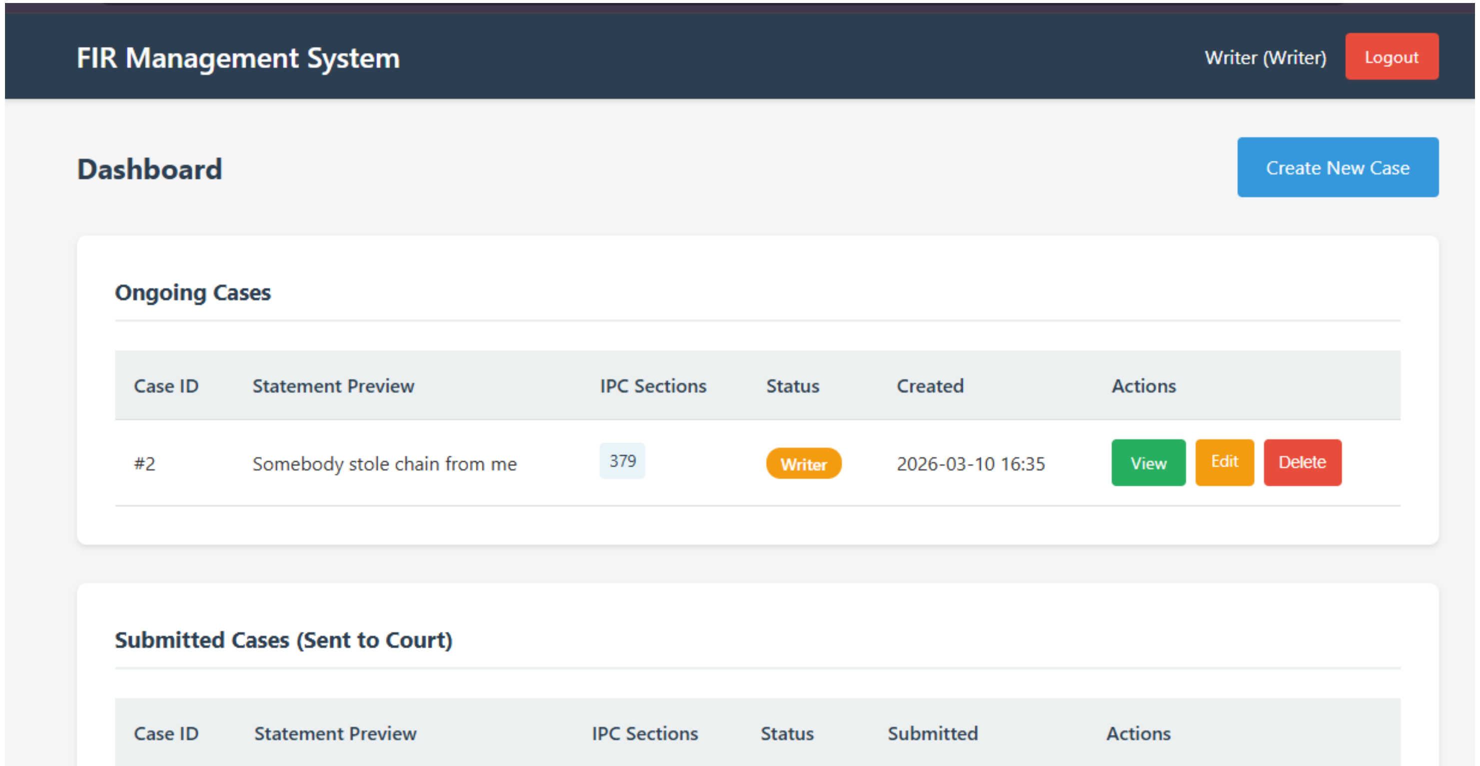
**2. Speech-to-Text Algorithm**
- Converts voice input into text
- Enables voice-based complaint filing

**3. OCR Algorithm**
- Extracts text from images/documents
- Supports document-based complaints

**4. FIR Draft Generation Algorithm**
- Organizes extracted data
- Generates structured FIR document
- Maps to relevant legal sections

# Experimental Results



# Experimental Results

**Case #2**

**Statement**

Somebody stole chain from me

**Allocated IPC Sections**

**IPC Section 379**

**Case Information**

| | |
|---|---|
| **Input Type:** | Text |

**Case Information**

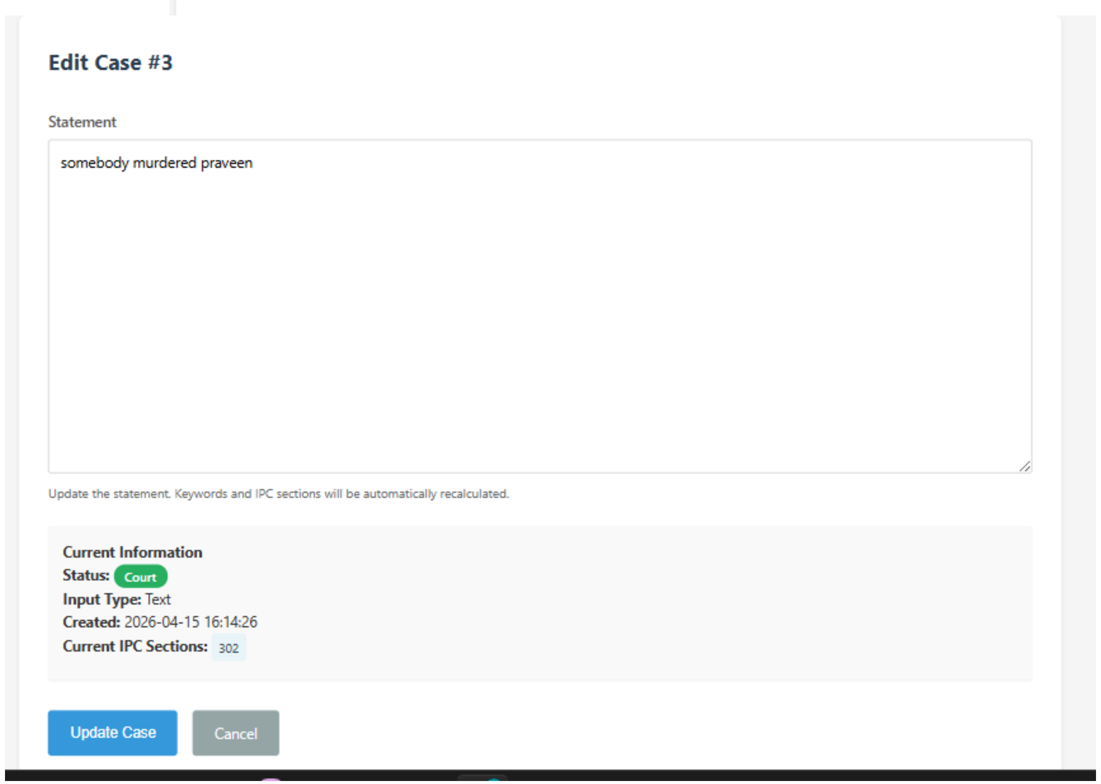
| | |
|---|---|
| **Input Type:** | Text |
| **Aadhaar Number:** | Not provided |
| **Created By:** | Writer |
| **Created At:** | 2026-03-10 16:35:15 |

**Evidence Inventory (Visible to All Station Personnel)**

*No evidence added yet.*

Evidence Description

# Experimental Results




## First Information Report

Case #3

| | |
|---|---|
| **Case ID** | 3 |
| **Status** | Court |
| **Input Type** | Text |
| **Created By** | Writer |
| **Created At** | 2026-04-15 16:14:26 |

### *Statement*

somebody murdered praveen

### *Allocated IPC Sections*

- IPC Section 302

# Comparison with Existing Systems

- The traditional manual FIR system is time-consuming, paper-based, and highly dependent on police officers, leading to delays and increased chances of human errors. It also requires citizens to have basic legal knowledge, making the process difficult for many users.
- Existing online FIR systems improve accessibility by allowing digital submission, but they still lack intelligent features such as AI-based guidance, automatic crime classification, and support for voice or document-based inputs.
- The proposed E-FIR Hub system overcomes these limitations by integrating AI technologies like NLP, Speech-to-Text, and OCR. It enables multi-format complaint submission, automatically classifies crimes, suggests relevant legal sections, and generates structured FIR drafts.
- Overall, the proposed system provides faster processing, improved accuracy, reduced manual effort, and a more user-friendly and efficient FIR registration process compared to existing systems.

# Testing Coverage

- Functional testing was performed to ensure that all modules, including user input, AI processing, FIR generation, and verification, are working correctly without errors.
- Input testing was conducted using different formats such as text, voice, and document inputs to verify the system’s ability to handle multi-input data effectively.
- Integration testing was carried out to check the interaction between modules like Speech-to-Text, OCR, NLP processing, and FIR generation for smooth workflow execution.
- Validation testing was done to evaluate crime classification accuracy, correctness of legal section mapping, and reliability of the generated FIR drafts, ensuring overall system performance and consistency.

# Deployment Status

- The system is currently developed and tested in a local development environment.
- Core modules such as data processing, NLP-based classification, FIR generation, and verification have been successfully implemented.
- Integration of Speech-to-Text and OCR functionalities has been completed and tested.
- Full-scale deployment on cloud/server is planned for the next phase of the project.
- Future work includes hosting the application, improving scalability, and enabling real-time user access.

# Conclusion

- The E-FIR Hub system successfully addresses the limitations of traditional FIR registration by introducing an AI-based approach that simplifies and automates the process.
- The integration of technologies such as NLP, Speech-to-Text, and OCR enables the system to handle multi-format inputs and accurately analyze complaint data.
- The system effectively classifies crimes, suggests relevant legal sections, and generates structured FIR drafts, reducing manual effort and minimizing errors.
- Overall, the project improves efficiency, accessibility, and reliability in FIR registration, making it more user-friendly for citizens and supportive for law enforcement agencies.

# THANK YOU